UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	PRE CALCULUS
COURSE CODE	MAT133
EXAMINATION	OCTOBER 2026
TIME	2 HOURS

NSTRUCTIONS TO CANDIDATES

1. This question paper consists of 5 questions.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) a one – page Appendix 1
   iii) an Answer Booklet - provided by the Faculty
   iv) a graph paper - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 9 printed pages

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QUESTION 1

a) Solve the inequality 6x - 7 > 4. Express the answer in interval notation.

(4 marks)



Figure: solution on a number line.

b) Solve the inequality |1 - 2x| > 3. Express the answer in interval notation and on a number line.

(5 marks)

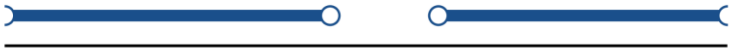


Figure: solution on a number line.

c) Solve the equation |5x + 4| = 3.

(4 marks)

QUESTION 2

a) Sketch $y = (x - 1)^2 - 3$ for $-2 \le x \le 4$. State the domain and range.

(4 marks)

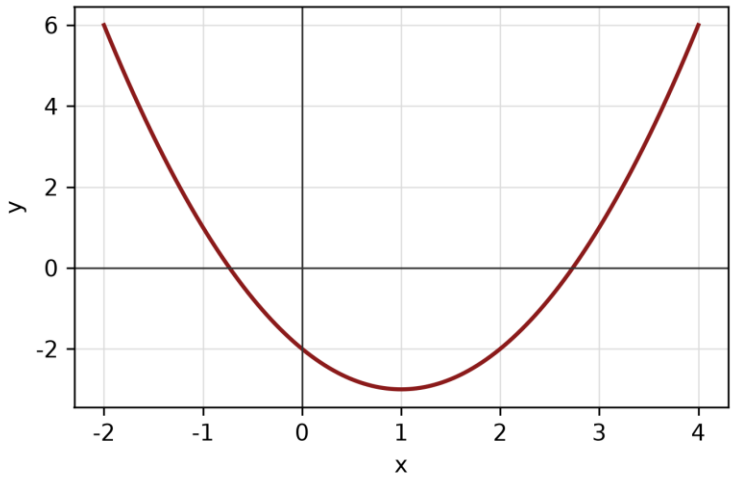


Figure: function graph.

b) Given f(x) = 4x + 3 and g(x) = 3x + 1, find (f + g)(x), (f - g)(x) and (f g)(x). State the domain of each.

(4 marks)

c) The endpoints of a diameter of a circle are (6, 5) and (-1, 4). Find the centre, radius and equation of the circle.

(5 marks)

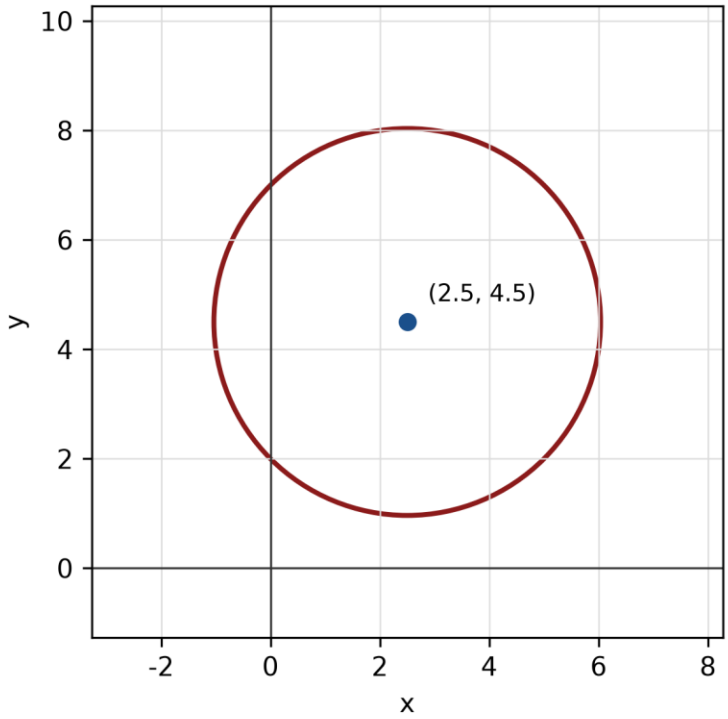


Figure: circle from diameter.

QUESTION 3

a) Given $z_1 = -4 + 4i$ and $z_2 = 2 - 3i$, express z_1/z_2 in the form a + bi.

(5 marks)

b) Find the modulus and argument of z = 8 + 15i. Give the argument in degrees correct to one decimal place.

(5 marks)

c) Use long division to find the quotient and remainder when f(x) = x^3 + 5x² + 6x + 1 is divided by x - 1.

(5 marks)

QUESTION 4

a) Solve the simultaneous equations
x - y = 5
-3x + 2y = -12.

(4 marks)

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b) Graph the system of inequalities below and shade the feasible region. State the vertices of the region.
$x \geq 0, y \geq 0, 1x + 1y \leq 8.$

(5 marks)

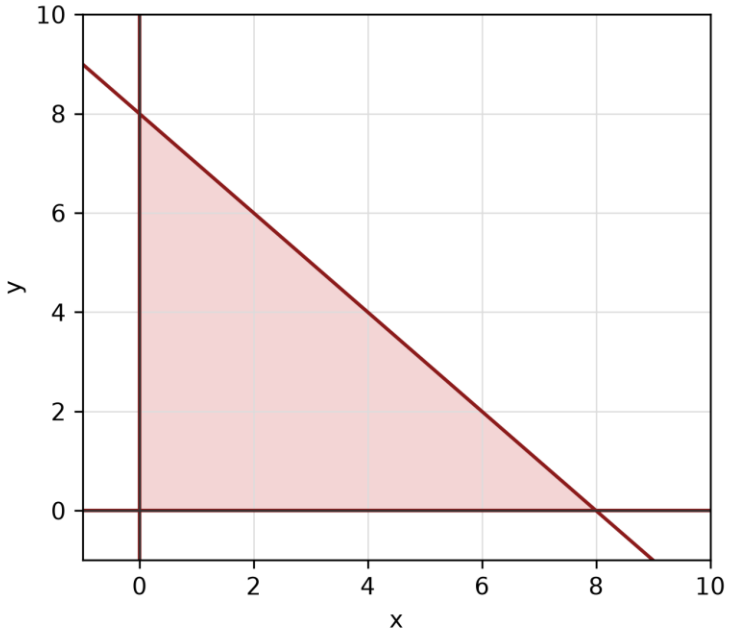


Figure: feasible region.

c) Sketch the circle $x^2 + 8x + y^2 + 10y + 37 = 0$. State its centre and radius.

(4 marks)

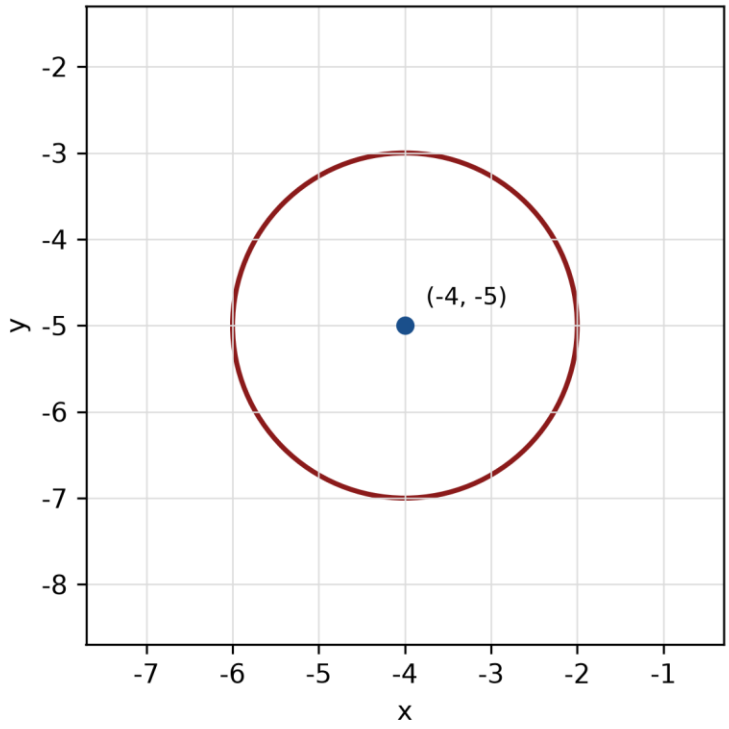


Figure: circle.

QUESTION 5

a) For $y = -3 \sin(2x + 0) + 2$, state the amplitude, period and phase shift, then sketch the graph

for -180° ≤ x ≤ 180°.

(5 marks)

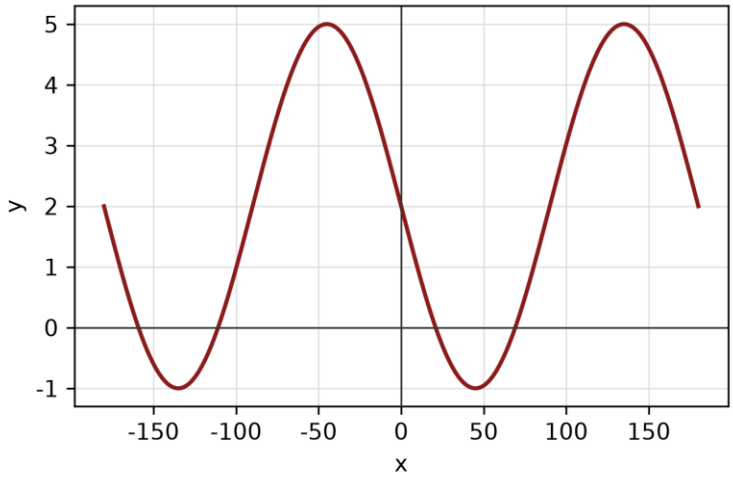


Figure: trigonometric graph.

b) Find the area of the sector of a circle with radius 8 cm and central angle 135°.

(4 marks)

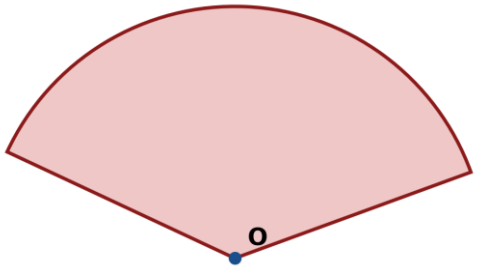
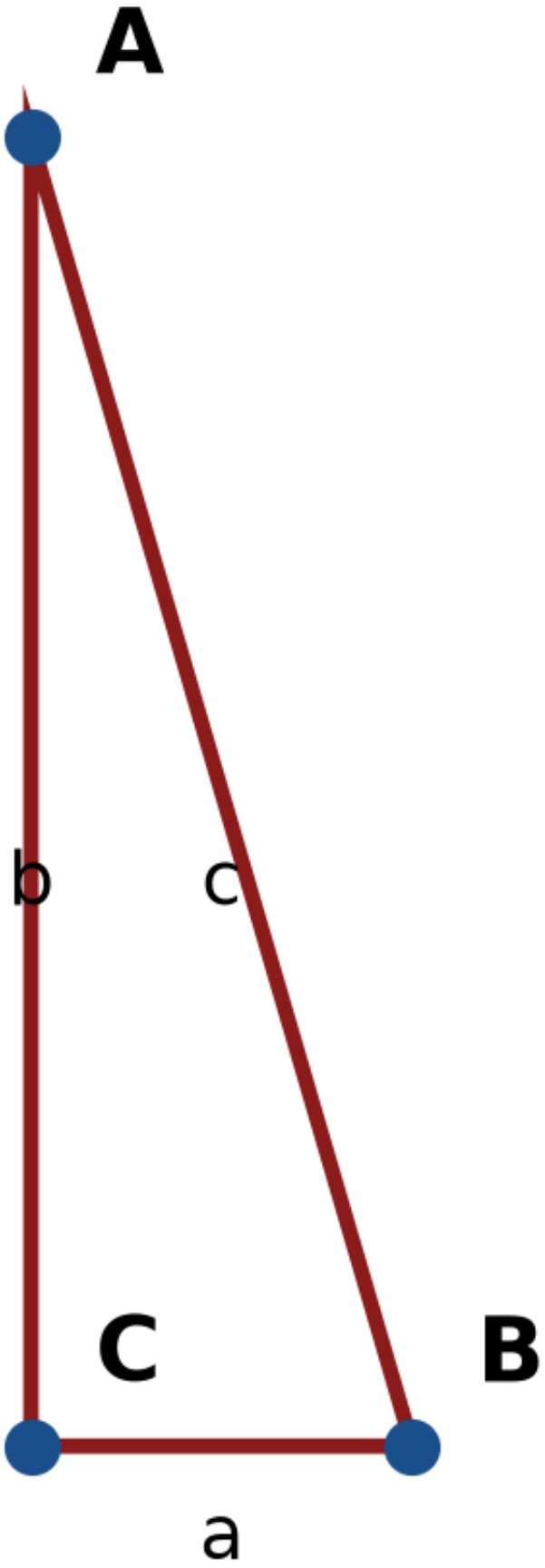


Figure: sector.

c) Solve the triangle with sides a = 14 cm, b = 48 cm and c = 50 cm. Find all three angles.
(5 marks)

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Figure: right triangle.

END OF QUESTION PAPER

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APPENDIX 1

Formula sheet (as supplied with the paper).

$\sin^2 A + \cos^2 A = 1$

$\sin(A + B) = \sin A \cos B + \cos A \sin B$

$\cos(A + B) = \cos A \cos B - \sin A \sin B$

$\tan(A - B) = (\tan A - \tan B) / (1 + \tan A \tan B)$

$\cos 2A = \cos^2 A - \sin^2 A$

$\cos 2A = 1 - 2\sin^2 A = 2\cos^2 A - 1$

$\sin 2A = 2 \sin A \cos A$

$\tan^2 A + 1 = \sec^2 A$

$\tan 2A = 2 \tan A / (1 - \tan^2 A)$

Arc length, $s = r\theta$

Area of sector = $\frac{1}{2}r^2\theta$

$a / \sin A = b / \sin B = c / \sin C$

$a^2 = b^2 + c^2 - 2bc \cos A$

Area of triangle = $\frac{1}{2}ab \sin C$

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